

Marine molecular biogeochemist — physical transport to microbial rates, on particles and in surface sediments

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STATEMENT I turn marine measurements into rates — **single-particle imaging into carbon-flux terms**, **^{13}C label into compound-specific rates**, **porewater and redox gradients into microbial turnover**. Trained in **^{15}N rate measurement** (isotope pairing, MIMS, pool dilution) for nitrification, denitrification, anammox, DNRA, N_2 fixation and N_2O — the toolkit HypoWaves turns on episodic hypoxia.

01 EDUCATION

- 2020–2026** **M.Sc. Marine Geosciences**, University of Bremen — defended **23.03.2026** · *Environmental Archives Projects 1.0* · *Project Studies and Scientific Training 2.0* · overall 2.3 “gut” (German scale, 1.0 best; transcripts enclosed) · examiners Prof. Dr. K.-U. Hinrichs, Prof. Dr. J.-H. Hehemann
- 2016–2018** **Doctoral research, Hydrogeochemistry**, Ruhr-Universität Bochum — two years of independent research on trace elements in groundwater and drinking-water treatment (**PHREEQC speciation of 1,211 analyses**; coordination of a four-well study with a **120-sample plan over 0–185 m**); discontinued when the supervisor retired on health grounds
- 2008–2011** **M.Sc. Geosciences**, Ruhr-Universität Bochum — thesis *very good* (80/100), overall *good* (70/100)
- 2002–2006** **B.Sc. Geophysical Engineering**, Kocaeli University

02 RESEARCH EXPERIENCE

- M.SC. THESIS**
10/2024–12/2025
Instant Visualization of Organic Matter Transformation in the Ocean
MARUM (Hinrichs Lab) · MPI for Marine Microbiology, Marine Glycobiology · BIOM (Hehemann Lab) · defended 23.03.2026 · DFG TRR 420 CONCENTRATE · Guest Scientist, AG Hehemann; Research Student, MPI Bremen
- Designed and ran **five particle-degradation experiments (ADE I–V) on six marine *Vibrio* strains** — mono- and co-culture, biological triplicates, incubations to 590 h — tracking **302–353 individually matched particles** per run. Broadcasters removed **78.1 ± 17.6 %** of particle volume against aggregators' **50.5 ± 24.2 %** ($t = 7.03$, $P = 1.7 \times 10^{-10}$).
 - Phosphate acts as a specific metabolic switch**: at 0.2 mM the broadcaster phenotype removed **2.1× more particle volume** (median 82.5 % vs 39.3 %; $U = 959$, $P = 0.0005$); the advantage vanishes under depletion, and in co-culture the 2.2× advantage **collapses to 1.05×** — traced to one alginate-lyase gene (**PL7G**), not the phenotype. Built the system from scratch: **calcium-alginate microparticles** ($124.0 \pm 49.1 \mu\text{m}$) and a custom **Marine Minimal Medium family** isolating one nutrient per run.
 - Built the **particle-imaging-to-rate pipeline** in **Python 3.11**: time-lapse microscopy → ImageJ Feret/ESD sizing → particle volume → daily carbon degradation rate → carbon loss per metre, with **Hungarian-algorithm matching** — a chain that transfers directly to in-situ particle-camera imagery. A settling model on the measured rates **predicts** Martin exponents of 0.95 vs 0.78 (~11 % vs ~17 % of particle carbon reaching 1,000 m).
- PROJECT**
2022
 ^{13}C stable-isotope probing and diffusive-flux geochemistry — Helgoland Mud Area, North Sea
Biogeochemistry Projects, MARUM / AWI · core HE595 GC-11, 452 cm · support M. Elvert, V. Heuer, L. Wörmer; porewater section reviewed by Prof. Dr. S. Kasten
- Designed and ran **^{13}C -acetate incubations in three arms** (48 h/28 °C, 120 h/28 °C, 120 h/8 °C, frozen t_0) on four depths spanning the **sulfate-reduction, sulfate–methane-transition and methanogenic zones**, anaerobic throughout. Traced the label into **individual fatty acids by GC-C-IRMS** ($\text{C}_{16}:1\omega 7$, $\text{C}_{16}:1\omega 5$): **bacterial communities respond to elevated temperature faster than archaeal**, and the OH-archaeol/archaeol ratio resolves an **ANME-1 → ANME-2 shift** across the transition.
 - Extracted porewater by **rhizone sampling (0.1 μm)**, measured nutrients and DIC on a **QuAAtro39**, and converted the sulfate gradient into a diffusive flux by **Fick's law with tortuosity correction** ($\phi = 0.6$, $\theta^2 = 2.02$): **$J = -0.219 \text{ mol m}^{-2} \text{ yr}^{-1}$** . Documented the **redox cascade after oxygen exposure** of reduced sediment — sulfide re-oxidation, Fe and Mn re-precipitation, phosphate re-adsorption onto fresh oxides.

EMPLOYED 05/2022– 01/2023	Student Assistant, Organic Geochemistry Laboratories — Alfred Wegener Institute Employer-attested: ultrasonication extraction · open-column liquid chromatography · silylation of phytosterols · GC-MS and HPLC-MS of hydrocarbon and polar fractions · identification and quantification of HBIs and GDGTs (GC-FID 7890A; HPLC-APCI-MS 1260/6120). Material: IODP Exp. 374 (Ross Sea), PS81, PS119; POSY network.
PROJECT 2023– 09/2024	Chemical imaging of a millimetre-scale redox gradient — hypersaline microbial mat MARUM, AG Hinrichs · Dr. B. Nettersheim, with Prof. K.-U. Hinrichs and Y. Zander Developed extraction-free MALDI-FT-ICR-MS imaging (Bruker solarix XR 7 T, 50 µm); discovered PC-DEG C44:3 and 14 associated new biomarkers for sulfate-reducing bacteria. Resolved eleven biomarker-defined redox zones across two mats, integrated with µ-XRF mapping and O₂, H₂S and pH microsensor profiles I evaluated — oxic–anoxic transition at 1.35 mm.
PAPER 2025	Black Sea palaeo-biogeochemistry — single-authored multi-proxy manuscript Marine Environmental Archives, U Bremen · cores GeoB 7608-1/7609-1 · biomarkers at AWI Reconstructed the limnic–marine transition at 8.5–8.0 ka as a gradual ~500-year reconnection from a 13-horizon GDGT–alkenone dataset (TEX₈₆^H, RI-OH, U_K³⁷; BIT 0.30 → 0.12), with Bayesian SST calibration (BAYSPAR v1.1) integrated with XRF scanning, dinocyst palynology and greigite magnetics — sulfate influx driving sulfate reduction and iron-sulfide authigenesis.

03 RATE & TRANSPORT TRAINING

- Benthic flux measurement and sensors** (Dr. M. Holtappels, 2020) — microsensors, optodes and planar optodes, chamber incubations, rate integration, **eddy correlation**; DOU vs TOU.
- Rates from radiotracer and porewater profiles** (Dr. G. Wegener, 2021) — **¹⁴CH₄ oxidation and ³⁵SO₄²⁻ reduction rates** (after Kallmeyer et al.), SMTZ flux estimates.
- GLOMAR “Stable Isotopes in Ecology”** (Dr. H. Marchant, Dr. M. Elvert, 2022) — **¹⁵N rate measurement: isotope pairing, MIMS, pool-dilution and bottle-effect corrections** for nitrification, denitrification, anammox, DNRA, N₂ fixation and N₂O; NanoSIMS and CARD-FISH. **Biogeochemistry I–II and Marine Environmental Archives over nine semesters** (1.0), U Bremen.

04 FIELD & TECHNICAL CAMPAIGNS

- Co-organised the two-week geothermal field campaign, Puna/Altiplano, Argentina** (RUB + TU Darmstadt, 2018) — supervised 20+ students sampling geothermal fields and mud volcanoes.
- Geophysical borehole logging at 12 geothermal wells** (full-waveform sonic, acoustic televiewer, gamma ray; GZB Bochum, 2016) · **GESEP School Scientific Drilling** (2018) · **IODP/ECORD core description school**, Bremen Core Repository (2022).

05 METHODS

ISOTOPES & INCUBATIONS	¹³ C-SIP design (anaerobic, multi-temperature) · ¹⁵N rate methods — isotope pairing, MIMS, pool dilution (course level) · GC-C-IRMS · lipid extraction, saponification/FAME, silylation, open-column LC
CHROMATOGRAPHY & MS	GC-MS · GC-FID (7890A) · HPLC-ESI/APCI-MS (1260/6120) · HBI and GDGT quantification · MALDI/LDI-FT-ICR-MS imaging (solarix XR 7 T)
TRANSPORT, RATES & CULTURE	diffusive flux from porewater gradients (Fick’s law, tortuosity) · radiotracer rate arithmetic (¹⁴ C, ³⁵ S) · settling and flux-attenuation modelling · PHREEQC · marine <i>Vibrio</i> mono- and co-culture, custom minimal-media design
DATA	Python 3.11 (NumPy, SciPy, pandas, matplotlib; examined course) · MATLAB / GNU Octave · BAYSPAR · ImageJ · Hungarian-algorithm tracking · Cytoscape

06 PUBLICATIONS & OUTPUTS

- 2026** **M.Sc. thesis**, defended 23.03.2026 (*enclosed as the self-written scientific text*)
- 2025** **Kandemiroglu, Gu & Hehemann: *Instant Visualization of Polysaccharide Degradation by Marine Bacteria***. Poster, Ocean Floor Symposium
- 2012** Wisotzky, F., **Kandemiroglu, O. C.** & Plassmann, C.: Nickel im Grundwasser. *gwf Wasser|Abwasser* **153**(7), 828–832 (*peer-reviewed*)

07 TEACHING, AWARDS & SERVICE

- 2016–18** **Four lecture courses over four semesters**, RUB — hydrochemistry, hydrogeology, geology · **Teaching Assistantship, RUB Research School**
- 2023–26** **Studienlotse** (student adviser), U Bremen · elected deputy, **Fachbereichsrat FB 5** (2024–26)
- 2021** Editor and compiler, M.Sc. Master Conference abstracts, U Bremen (~25 authors)
- 2010** **DAAD STIBET grant**, RUB International Office

08 REFERENCES two referees, as requested

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